

**GRAPH THEORY:  
HOMEWORK 4  
BOOK SECTION 2.1.**

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**1.** Prove that every tree with maximum degree  $\Delta > 1$  has at least  $\Delta$  vertices of degree 1. Show that this is the best possible by constructing an  $n$ -vertex tree with exactly  $\Delta$  leaves, for each choice of  $n, \Delta$  with  $n > \Delta \geq 2$ .

*Proof.* □

**2.** Prove that every graph with diameter  $d$  has an independent set of size at least  $\lceil (1+d)/2 \rceil$ .

*Proof.* □

**3.** Let  $d_1, \dots, d_n$  be positive integers with  $n \geq 2$ . Prove that there exists a tree with vertex degrees  $d_1, \dots, d_n$  if and only if  $\sum d_i = 2n - 2$ .

*Proof.* □

**4.**

- (a) Prove that every tree is bipartite.
- (b) Prove that every tree  $T$  has a leaf in its larger partite set (in both if they have equal size).

*Proof.* □

**5.**

- (a) Prove that the distance function  $d(u, v)$  on pairs of vertices of a graph satisfies the *triangle inequality*:  $d(u, v) + d(v, w) \geq d(u, w)$ .
- (b) Use (a) to prove that  $\text{diam}(G) \leq 2 \cdot \text{rad}(G)$  for every graph  $G$ .
- (c) For all positive integers  $r$  and  $d$  that satisfy  $r \leq d \leq 2r$ , construct a simple graph with radius  $r$  and diameter  $d$ . (Hint: Build a suitable graph with one cycle.)

*Proof.* □